

Electronic Health Record: Pharmacy Organization Module

Using Hyperledger Fabric and Docker Swarm

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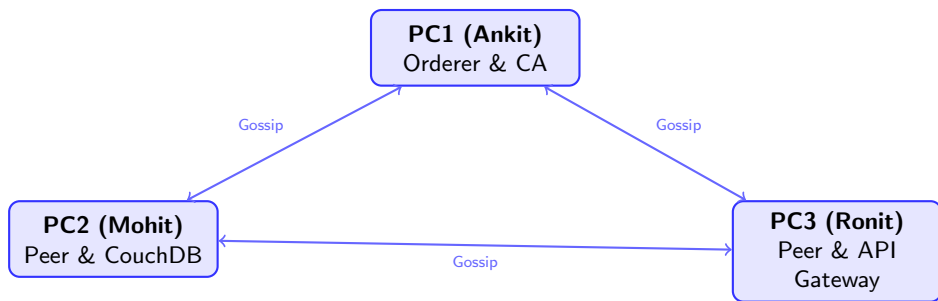
Outline

- 1 Introduction
- 2 System Architecture
- 3 Individual Contributions
 - Ankit Chaudhari
 - Mohit Tomar
 - Ronitsingh Rawat
- 4 Verification
- 5 Conclusion

The Electronic Health Record (EHR) System is a decentralized, secure, and scalable platform designed for pharmaceutical inventory and patient record management.

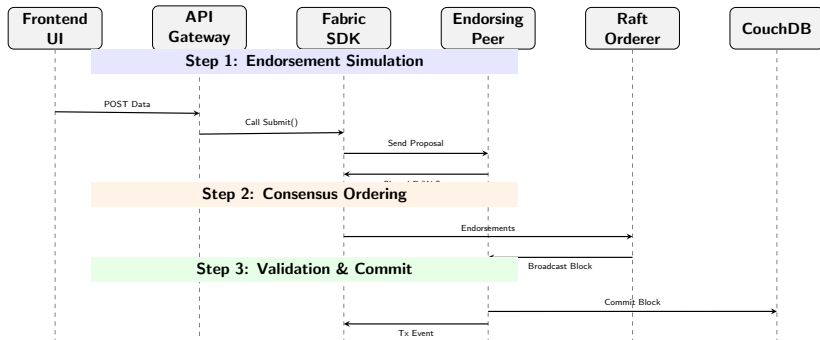
- **Decentralization:** Built on Hyperledger Fabric.
- **Privacy:** Attribute-Based Access Control (ABAC).
- **Scalability:** Docker Swarm orchestration across multiple nodes.
- **Hybrid Storage:** On-chain metadata and off-chain IPFS document storage.

High-Level Architecture



- Layered Tiers: Presentation, Application, Blockchain, Storage.
- Distributed over 3 physical machines via Tailscale VPN.

Transaction Flow: Write Operation

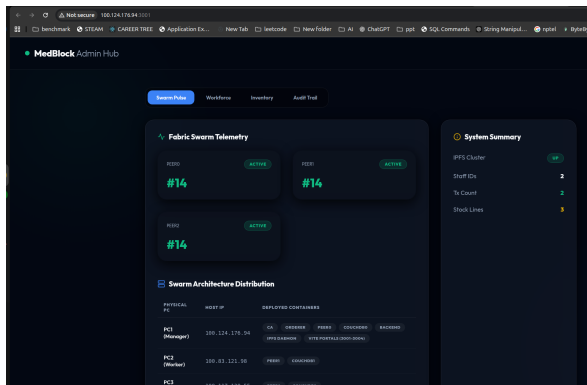


Role & Responsibility

Responsible for core network infrastructure, Docker Swarm orchestration, and the Manager Governance Portal.

- **Containers Hosted on PC1:** Fabric CA, Raft Orderer, Peer0, CouchDB0, Manager Portal.
- **Key Highlights:** Automated `deploy.sh` script, MSP distribution via SCP, and Network Health Monitoring.

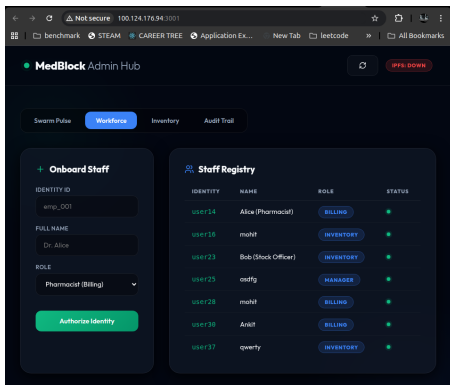
Manager Dashboard (Ankit Chaudhari - 25CSM2R03)



- **Description:** A high-level governance interface allowing hospital administrators to seamlessly oversee all interconnected system modules, evaluate real-time API gateways, and monitor global network health metrics.

Contribution by: Ankit Chaudhari (25CSM2R03)

Workforce Registry (Ankit Chaudhari - 25CSM2R03)



- **Description:** The primary employee enrollment interface. It securely assigns specific ABAC roles (Doctor, Pharmacist, Inventory) and automatically provisions deterministic cryptographic X.509 identities via the Fabric CA.

Contribution by: Ankit Chaudhari (25CSM2R03)

Network Monitoring (Ankit Chaudhari - 25CSM2R03)

```
ankit@ankit-Aspire-A715-75G:~/fabric-network/fabric-network-swarm$ bash scripts/swarm_status.sh

=== Swarm Nodes ===


| ID                        | HOSTNAME                                | STATUS | AVAILABILITY | MANAGER STATUS | ENGINE VERSION |
|---------------------------|-----------------------------------------|--------|--------------|----------------|----------------|
| k27mf7Bzf65l335xi8t5qbp * | ankit-Aspire-A715-75G                   | Ready  | Active       | Leader         | 28.2.2         |
| w9qzs7smzp6zh6yw86hprk017 | rajput-mt750XGK                         | Ready  | Active       |                | 28.5.2         |
| etmt48eh03hlpv8l1tnkyae5r | ronit-VivoBook-ASUSLaptop-X512FL-X512FL | Ready  | Active       |                | 28.2.2         |



=== Swarm Services ===


| ID           | NAME                     | MODE       | REPLICAS | IMAGE                          | PORTS                                |
|--------------|--------------------------|------------|----------|--------------------------------|--------------------------------------|
| ozpifaznhzpa | ehrswarm-ca_ca_org1      | replicated | 1/1      | hyperledger/fabric-ca:1.5      | *:7054->7054/tcp, *:17054->17054/tcp |
| kvyntcjbv4f  | ehrswarm-orderer_orderer | replicated | 1/1      | hyperledger/fabric-orderer:2.5 | *:7050->7050/tcp                     |
| ccyk1ysdmqz  | ehrswarm-peer0_couchdb0  | replicated | 1/1      | couchdb:3.3.2                  | *:7984->5984/tcp                     |
| lk9bojkab2sg | ehrswarm-peer0_peer0     | replicated | 1/1      | hyperledger/fabric-peer:2.5    | *:7051->7051/tcp                     |
| kvufuvfoalwf | ehrswarm-peer1_couchdb1  | replicated | 1/1      | couchdb:3.3.2                  | *:8984->5984/tcp                     |
| ng6o65oxsdqg | ehrswarm-peer1_peer1     | replicated | 1/1      | hyperledger/fabric-peer:2.5    | *:8051->8051/tcp                     |
| touyqx55y8tj | ehrswarm-peer2_couchdb2  | replicated | 1/1      | couchdb:3.3.2                  | *:9984->5984/tcp                     |
| tu3zicz4nzw  | ehrswarm-peer2_peer2     | replicated | 1/1      | hyperledger/fabric-peer:2.5    | *:9051->9051/tcp                     |



=== Overlay Networks ===

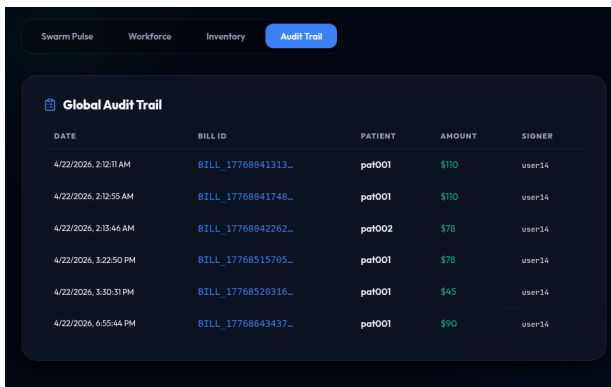

| NAME         | TYPE             | STATUS        |
|--------------|------------------|---------------|
| nnulxi14yyea | fabric-swarm-net | overlay swarm |


ankit@ankit-Aspire-A715-75G:~/fabric-network/fabric-network-swarm$
```

- **Description:** Provides real-time graphical visualization of overall Swarm node availability, current ledger block height across physical machines, and reliable Gossip data synchronization status.

Contribution by: Ankit Chaudhari (25CSM2R03)

Blockchain Audit Trail (Ankit Chaudhari - 25CSM2R03)



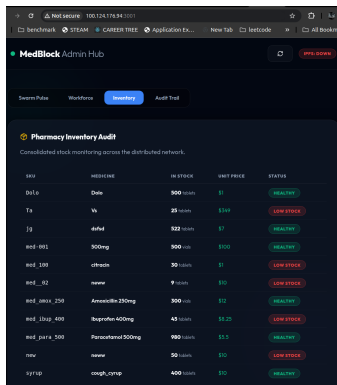
The screenshot displays a web application interface for a Blockchain Audit Trail. At the top, there are four tabs: 'Swarm Pulse', 'Workforce', 'Inventory', and 'Audit Trail', with 'Audit Trail' being the active tab. Below the tabs, the title 'Global Audit Trail' is shown with a document icon. The main content is a table with five columns: 'DATE', 'BILL ID', 'PATIENT', 'AMOUNT', and 'SIGNER'. The table contains six rows of transaction data, all recorded on 4/22/2026. The 'AMOUNT' column uses green text for positive values and red for negative values.

DATE	BILL ID	PATIENT	AMOUNT	SIGNER
4/22/2026, 2:12:11 AM	BILL_17768841313...	pat001	\$110	user14
4/22/2026, 2:12:55 AM	BILL_17768841748...	pat001	\$110	user14
4/22/2026, 2:13:46 AM	BILL_17768842262...	pat002	\$78	user14
4/22/2026, 3:22:50 PM	BILL_17768515785...	pat001	\$78	user14
4/22/2026, 3:30:31 PM	BILL_17768528316...	pat001	\$45	user14
4/22/2026, 6:55:44 PM	BILL_17768643437...	pat001	\$90	user14

- **Description:** An immutable and transparent record of all historical activities recorded on the ledger, effectively allowing hospital managers to easily pull global hospital records for comprehensive auditing and oversight.

Contribution by: Ankit Chaudhari (25CSM2R03)

Inventory Oversight (Ankit Chaudhari - 25CSM2R03)



The screenshot shows a web application titled 'MedBlock Admin Hub'. It has a navigation bar with 'Inventory' selected. Below the navigation bar, there's a section titled 'Pharmacy Inventory Audit' with the subtitle 'Consolidated stock monitoring across the distributed network.' Below this is a table with the following data:

SKU	MEDICINE	IN STOCK	UNIT PRICE	STATUS
D416	Dolo	900 tablets	\$1	HEALTHY
Ta	Va	25 tablets	\$349	LOW STOCK
Jg	dofol	522 tablets	\$7	HEALTHY
med_001	500mg	500 vials	\$100	HEALTHY
med_100	citracin	50 tablets	\$1	LOW STOCK
med_02	new	9 tablets	\$10	LOW STOCK
med_sku_250	Amoxicillin 250mg	300 vials	\$10	HEALTHY
med_ibup_400	Ibuprofen 400mg	45 tablets	\$0.25	LOW STOCK
med_para_500	Paracetamol 500mg	990 tablets	\$0.5	HEALTHY
new	new	50 tablets	\$10	LOW STOCK
syfcp	cough_syrup	400 tablets	\$10	HEALTHY

- **Description:** A specifically assigned read-only global oversight interface showing real-time medicine stock levels aggregated across all pharmaceutical points, seamlessly ensuring total transparency in supply.

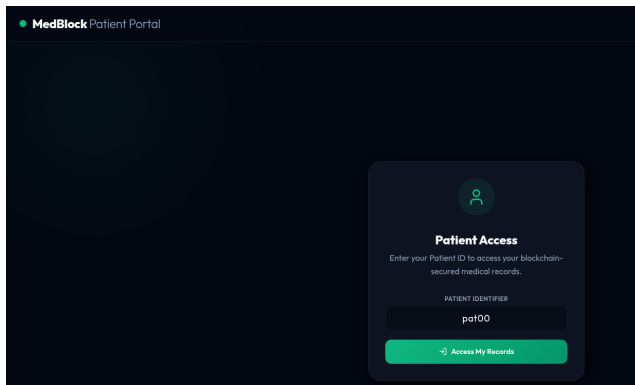
Contribution by: Ankit Chaudhari (25CSM2R03)

Role & Responsibility

Development of Node.js Chaincode, CouchDB Rich Queries, and clinical portals (Patient & Pharmacist).

- **Containers Hosted on PC2:** Peer1, CouchDB1, Patient Portal, Pharmacist Portal.
- **Key Highlights:** Identity Resolution algorithm, attribute-based access control, and IPFS anchoring logic.

Patient Portal (Part 1)



- **Description:** The core Self-Sovereign Identity (SSI) portal demonstrating patient access. The dashboard gives the patient a clear, chronological overview of their complete medical history logged safely onto the ledger.

Contribution by: Mohit Tomar (25CSM2R13)

Patient Portal (Part 2)

The screenshot displays a patient portal interface with a dark theme. At the top, there are three tabs: 'Medical History' (highlighted in blue), 'Access Governance', and 'Identity'. Below the tabs is a section titled 'Immutable Record Log' with a clock icon. A subtitle states: 'Your transactions are secured by Hyperledger Fabric and archived on IPFS.' Below this is a table with the following data:

DATE	BILL ID	AMOUNT	PROVIDER	IPFS	VIEW
4/23/2026	BILL_177695984...	\$110	user14	QhRE6hFTGg3zcV...	
4/23/2026	BILL_177696011...	\$100	user14	Qsc0M6Zm03Hc...	

- **Description:** Features decentralized off-chain IPFS document storage integration holding heavy medical reports, allowing patients to explicitly grant or revoke viewing rights selectively.

Contribution by: Mohit Tomar (25CSM2R13)

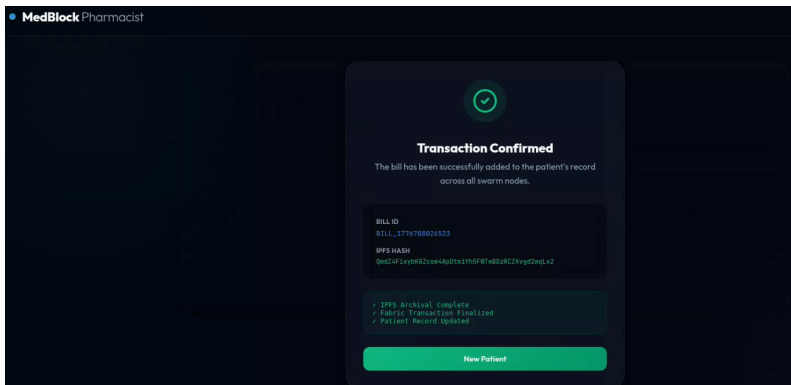
Pharmacist Portal (Part 1)

The screenshot shows the MedBlock Pharmacist Portal interface. At the top left, the logo 'MedBlock Pharmacist' is displayed. The main content area is divided into two panels. The left panel, titled 'Alice Wonderland' (Age: 24 - Verified Patient), shows 'Past Prescriptions' with a list item 'BILL_17767...' and a green price tag '\$110'. Below this is a button labeled '... New Search'. The right panel, titled 'Dispatch Bill & Prescription', contains a note: 'This transaction will be recorded on the Fabric ledger and archived on IPFS.' Below this, there is a 'BILLING AMOUNT (\$)' section with a text input field containing '110'. Underneath is a 'Medicines & Dosage' section with a text input field containing 'para' and a dropdown menu showing '10'. A button labeled '+ Add Medicine' is located below the dosage field. At the bottom of the right panel, a progress bar indicates 'Committing to Ledger...'.

- **Description:** A dedicated pharmacy billing interface operating purely on Attribute-Based Access Control (ABAC) principles, strictly limiting the pharmacist's view only to prescribed medications on demand.

Contribution by: Mohit Tomar (25CSM2R13)

Pharmacist Portal (Part 2)



- **Description:** Seamlessly tracks prescription fulfillment and features dynamic integration with the global supply chain ledger, allowing instant atomic updates bridging the billing flow and physical stock count.

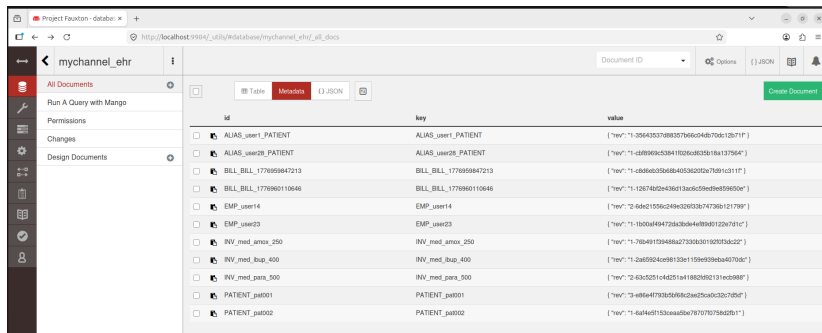
Contribution by: Mohit Tomar (25CSM2R13)

Role & Responsibility

Central API Gateway (Node.js SDK), Wallet Management, and Inventory Supply Chain Application.

- **Containers Hosted on PC3:** Peer2 (Redundancy), CouchDB2, API Gateway, Inventory Portal.
- **Key Highlights:** Smart Routing algorithm, deterministic certificate mapping, and high availability setup.

State Mirroring: CouchDB2 Interface



	id	key	value
☐	ALIAS_user1_PATIENT	ALIAS_user1_PATIENT	{ "rev": "1-39643537d88357b66c94db70dc12b71f" }
☐	ALIAS_user28_PATIENT	ALIAS_user28_PATIENT	{ "rev": "1-c8f8969c539415266c835b18a137564" }
☐	BILL BILL_1776959847213	BILL BILL_1776959847213	{ "rev": "1-c8d8e035b68b405362012e7b919c311f" }
☐	BILL BILL_1776960110646	BILL BILL_1776960110646	{ "rev": "1-12674b2e436d13ac6c59ed3e859650e" }
☐	EMP_user14	EMP_user14	{ "rev": "2-6de21596c249e3263b74736b121799" }
☐	EMP_user23	EMP_user23	{ "rev": "1-1b004f9472da3bd4e88d0122b7d1c" }
☐	INV_med_amox_250	INV_med_amox_250	{ "rev": "1-78b491f39488a273301c301901f3dc22" }
☐	INV_med_ibup_400	INV_med_ibup_400	{ "rev": "1-2a85924ce98133e1159e939eba4070dc" }
☐	INV_med_para_500	INV_med_para_500	{ "rev": "2-63c5251c4d251a41882b92131ecb888" }
☐	PATIENT_pat001	PATIENT_pat001	{ "rev": "3-88fe4f793b3b5b68c2a825ca0c32c7dbd" }
☐	PATIENT_pat002	PATIENT_pat002	{ "rev": "1-6a4e5f153ceaa50e787070756d2fb1" }

- **Description:** A live backend look at CouchDB world state mirroring, effectively proving perfect decentralized cross-node data availability, rich JSON querying capabilities, and consensus fidelity across the Swarm.

Contribution by: Ronit singh Rawat (25CSM2S06)

Inventory Supply Chain (Part 1)

The screenshot shows the MedBlock Inventory Management interface. At the top, there's a header with the MedBlock logo and the word 'Inventory'. Below this, there are two tabs: 'Stock Ledger' and 'Inventory Management', with the latter being the active tab. The main content area is divided into two panels. The left panel is titled '+ Add / Restock Medicine' and contains a form with the following fields: 'ITEM SKU / ID' (with the value 'med_001'), 'MEDICINE NAME' (with the value 'Paracetamol 500mg'), 'QUANTITY' (with the value '500'), 'BASE PRICE (\$)' (with the value '10.00'), and 'UNIT' (with a dropdown menu showing 'Tablets'). At the bottom of this panel is a green button labeled 'Update Stock Ledger'. The right panel is titled 'Inventory Compliance' and contains two sections: 'Blockchain Anchoring' with a description about generating a new block and ensuring zero 'ghost stock', and 'Auto-Deduction Active' with a description about linking pharmacy billing to the inventory.

MedBlock Inventory

Stock Ledger Inventory Management

+ Add / Restock Medicine

Updates are anchored to the blockchain and visible to pharmacists instantly.

ITEM SKU / ID
med_001

MEDICINE NAME
Paracetamol 500mg

QUANTITY
500

BASE PRICE (\$)
10.00

UNIT
Tablets

Update Stock Ledger

Inventory Compliance

Blockchain Anchoring
Every inventory update generates a new block on the ledger. This ensures zero "ghost stock" and complete auditability for controlled substances.

Auto-Deduction Active
Pharmacy billing is linked to this inventory. When a pharmacist issues a bill, these stock levels are automatically updated on the blockchain.

- **Description:** The primary entry point interface effectively responsible for smoothly adding bulk new medicine stock imports into the blockchain ledger, mathematically tied to irreversible, immutable timestamps.

Contribution by: Ronit singh Rawat (25CSM2S06)

Stock Management (Part 2)

The screenshot displays the MedBlock Inventory management interface. At the top, there's a navigation bar with the MedBlock logo and 'Inventory' text. Below this, a secondary bar contains 'Stock Ledger' and 'Inventory Management' tabs. The main content area is divided into two panels. The left panel, titled 'Pharmacy Real-Time Stock', shows a table of inventory items with columns for Item ID, Medicine, Quantity, Price, Last Sync, and Status. The right panel, titled 'Low Stock Alerts', displays a warning for 'Ibuprofen 400mg' with a level of 45.50.

ITEM ID	MEDICINE	QUANTITY	PRICE	LAST SYNC	STATUS
med_002	Iye	200 vials	\$50	4/21/2024, 9:45:09 PM	HEALTHY
med_amox_250	Amoxicillin 250mg	300 vials	\$12	4/21/2024, 9:45:12 PM	HEALTHY
med_ibup_400	Ibuprofen 400mg	45 tablets	\$8.25	4/21/2024, 9:45:34 PM	LOW STOCK
med_para_500	Paracetamol 500mg	980 tablets	\$5.5	4/21/2024, 9:45:30 PM	HEALTHY

- **Description:** Comprehensive visual list view managing all existing inventoried stocks and tracking automated ledger deductions, securely linked atomically to successful pharmacy billing transactions across the platform.

Contribution by: Ronit singh Rawat (25CSM2S06)

- **Ledger Sync:** All three peers (Peer0, Peer1, Peer2) reported identical block heights and matching hash values.
- **State Replication:** CouchDB instances on all nodes show synchronized world state.
- **Fault Tolerance:** Verified by taking one peer offline (PC2) and routing transactions through the redundant node (PC3).

Conclusion

- ① Successfully implemented a distributed EHR system across multiple physical nodes.
- ② Achieved perfect data consistency via Hyperledger Fabric's Gossip protocol.
- ③ Ensured security and privacy through ABAC and X.509 digital identities.
- ④ Built functional clinical, administrative, and inventory portals.

- **AI/ML Integration:** Implementing predictive analytics for hospital inventory resupply based on historical prescription rates.
- **Cross-Hospital Interoperability:** Expanding the channel configuration to allow external hospital organizations to query authorized records securely.
- **Mobile Application:** Developing dedicated native mobile applications for the Patient and Pharmacist portals for improved accessibility.

Thank You!